Variational Autoencoder

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PDL

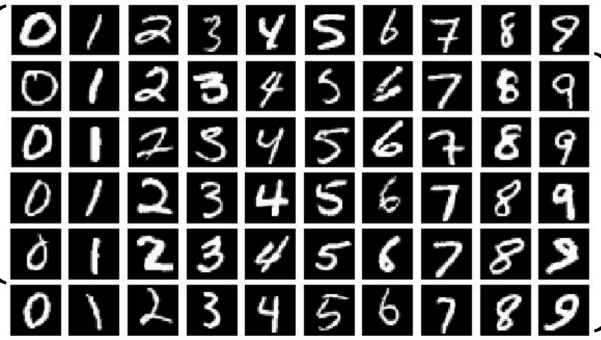
12th April, 2021

1. Varitional Auto-Encoder
   1.1. Generative Model
   1.2. VAE

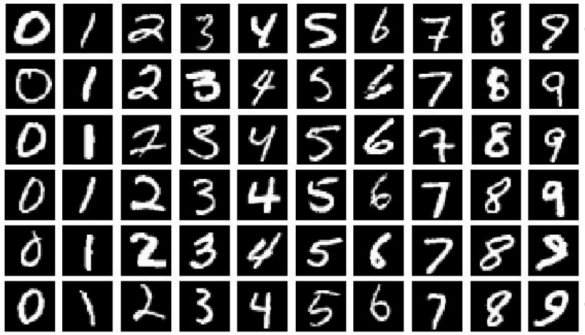
Generative model: overview

Generative Models:

- Model that describe the generating process of all observations
- Technically, they specify $P(x)$ (unsupervised) or $P(x, y)$ (supervised)

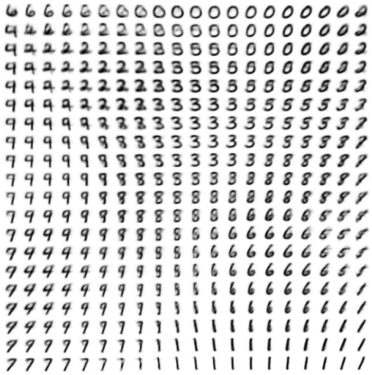
$\{x^{(n)}\} =$  $\sim p(x)$

$\{x^{(n)}, y^{(n)}\} =$ $y^{(n)} =$ "0" "1" "2" "3" "4" "5" "6" "7" "8" "9"

$x^{(n)} =$  $\sim p(x, y)$

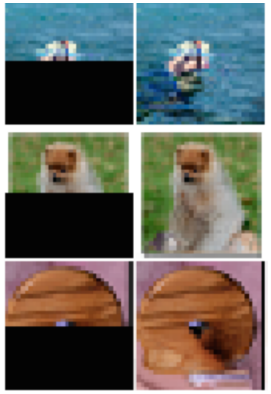
What can generative models do?

1. Generate new data.



Generation $p(x)$ [KW14]

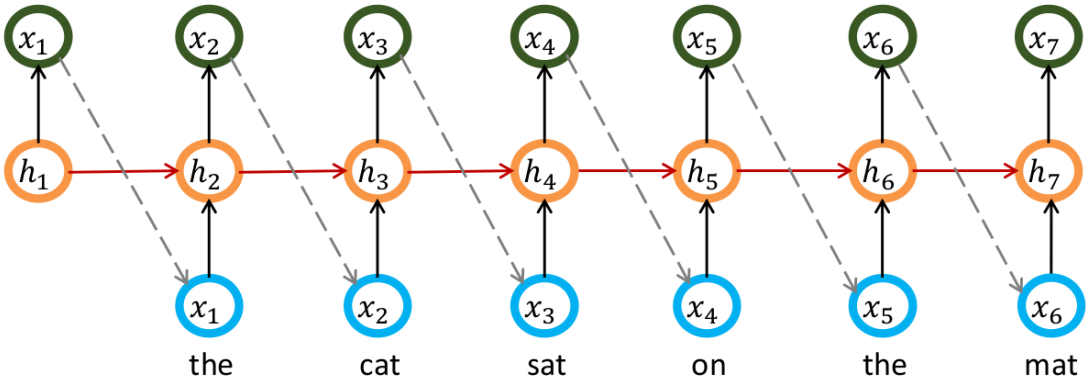
Conditional Generation $p(x|y)$ [LWZZ18]



Missing Value Imputation (Completion) $p(x_{\text{hidden}}|x_{\text{observed}})$ [OKK16]

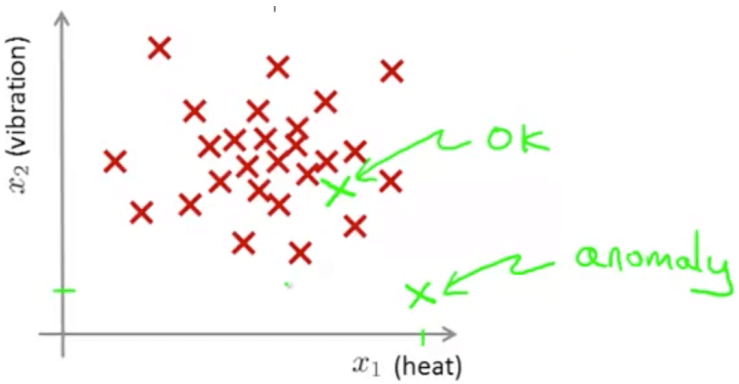
"the cat sat on the mat" $\sim p(x)$: Language Model.

$p(x_1 = \text{the})$ $p(\text{cat}|x_1)$ $p(\text{sat}|x_{1\ldots2})$ $p(\text{on}|x_{1\ldots3})$ $p(\text{the}|x_{1\ldots4})$ $p(\text{mat}|x_{1\ldots5})$ $p(\text{</s>}|x_{1\ldots6})$

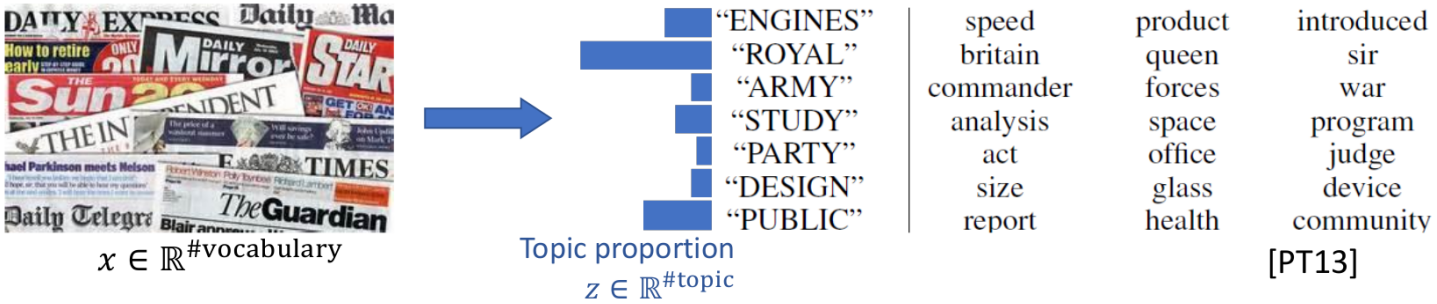


What can generative models do?

2. Density estimation $p(x)$



3. Draw semantic representation of data x (via latent variable z)



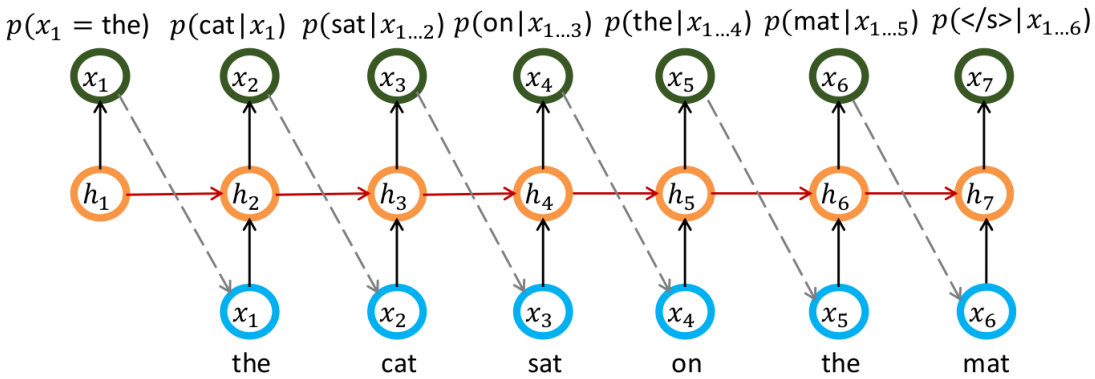
[PT13]

4. Supervised Learning: $\arg\max_{y} p(y, x)$

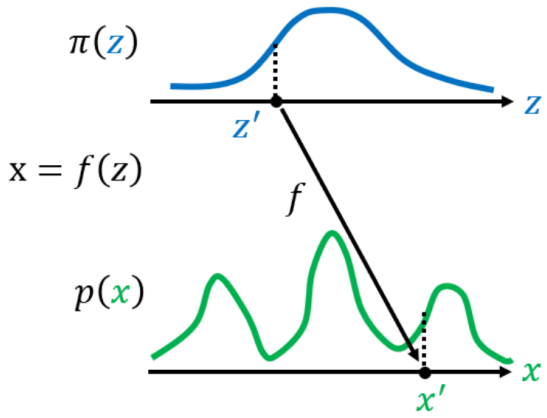
Taxonomy

1. Plain Generative Models: Directly model $P(x)$, no latent variable

"the cat sat on the mat" $\sim p(x)$: Language Model.

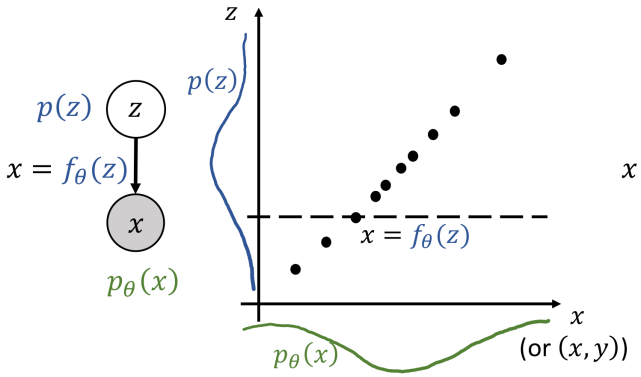


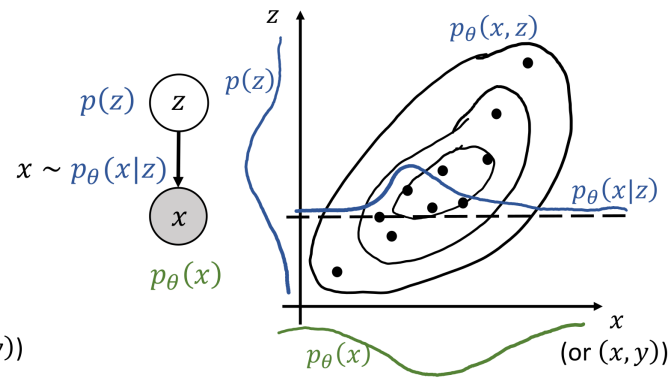
2. Latent Variable Models:
   - Map a simple distribution $z \sim \pi(z)$ to target complex distribution $x \sim p(x)$



Latent variable models

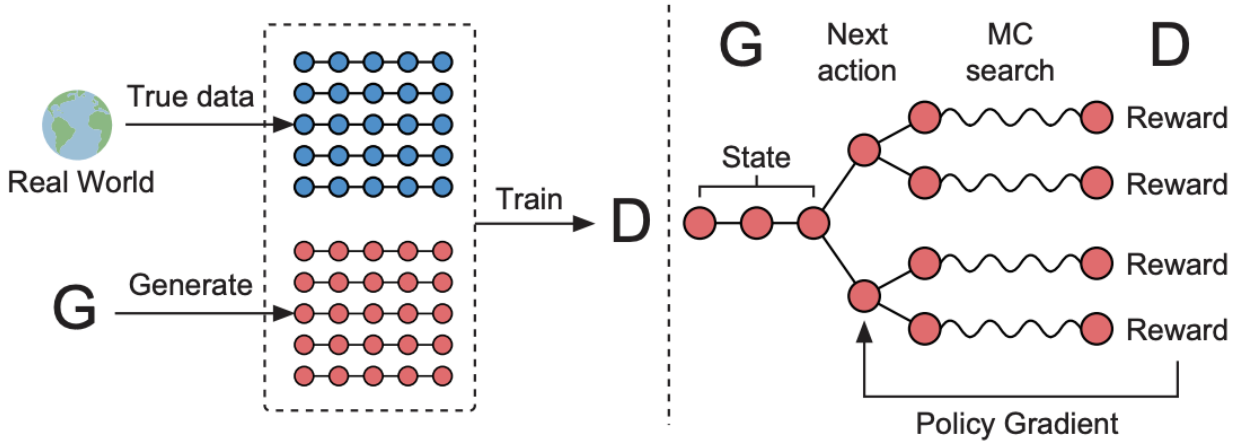
- Deterministic Generative Models
  - Dependency between x and z is deterministic: $x = f(z; \theta)$
  - e.g. GAN, Flow-based model
- Bayesian Generative Models
  - Dependency between x and z is probabilistic: $(x, z) \sim p(x, z; \theta)$
  - e.g. **VAE**, Bayesian Network



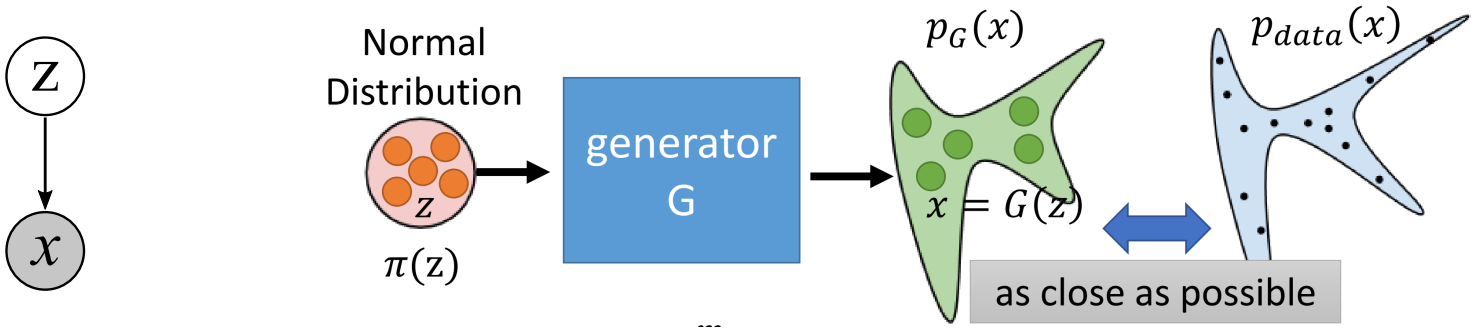


Why not GAN (in text)?

The text sequence is **discrete**. Gradient can not be back-propagated from discriminator to generator.



Preliminaries of VAE

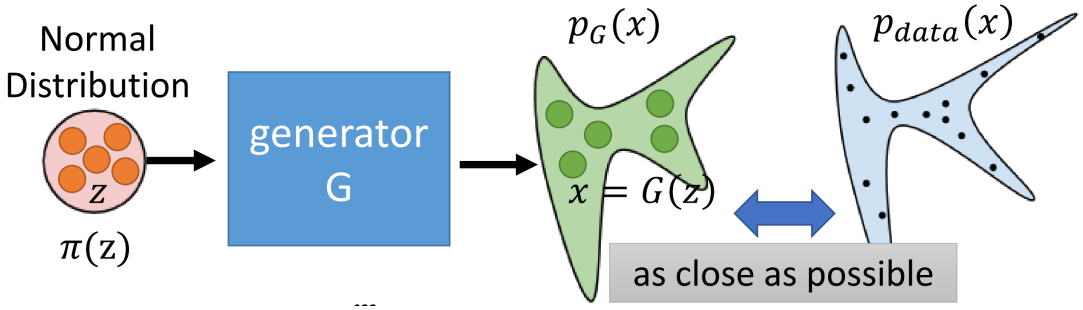


1. Assume training data x is generated from latent representation z
   - First, sample from true prior $z \sim \pi(z)$
   - Second, sample from conditional $x \sim P(x|z)$
2. Choose prior $\pi(z)$ to be simple, e.g. $\mathcal{N}(0, I)$
3. Conditional $P(x|z)$ is complex, represent with a generator G: $p(x|z;\theta)$

Setting up the objective

$\pi(z) = \mathcal{N}(0, I)$

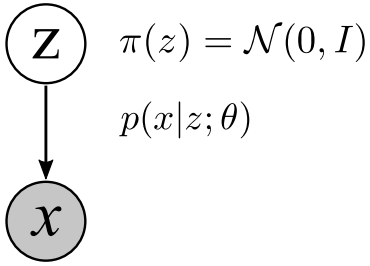
$p(x|z;\theta)$



We are aiming maximize the likelihood of each $x_1, \cdots, x_m \in X$

$$P(x_i) = \int p(x_i|z;\theta)\pi(z)dz$$

Setting up the objective



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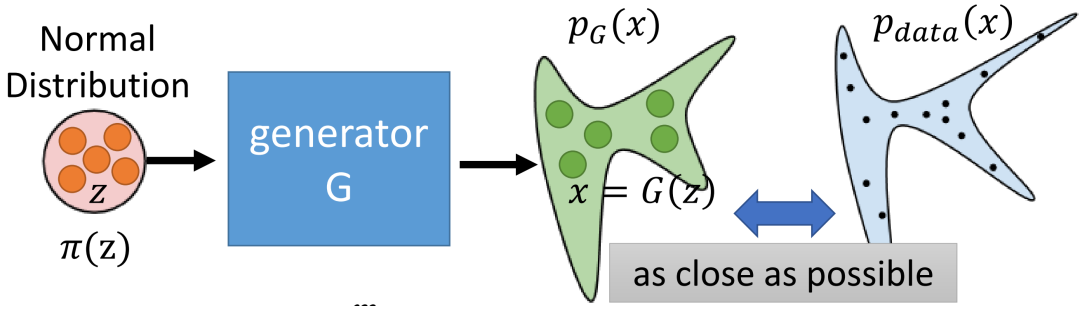

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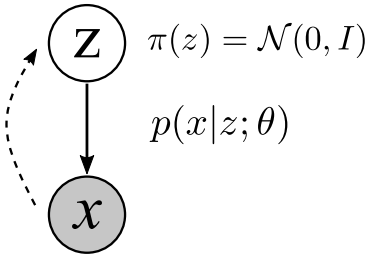
$$P(x_i) = \int p(x_i|z;\theta)\pi(z)dz$$

- In practice, for most z, $P(x_i|z)$ will be nearly zero, and contribute almost nothing to estimate of $P(x_i)$.
- We need sample values of z that are likely to have produced x_i, and compute $P(x_i)$ just from those

Posterior of latent z



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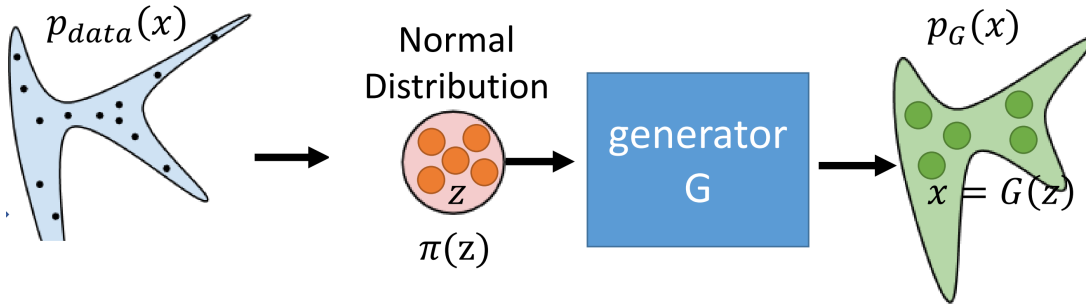
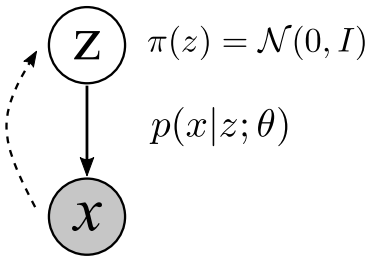
$p(x|z;\theta)$



Posterior density of z:

$$P(z|x_i) = \frac{P(x_i|z)P(z)}{P(x_i)} = \frac{p(x_i|z;\theta)\pi(z)}{P(x_i)}$$

Posterior of latent z



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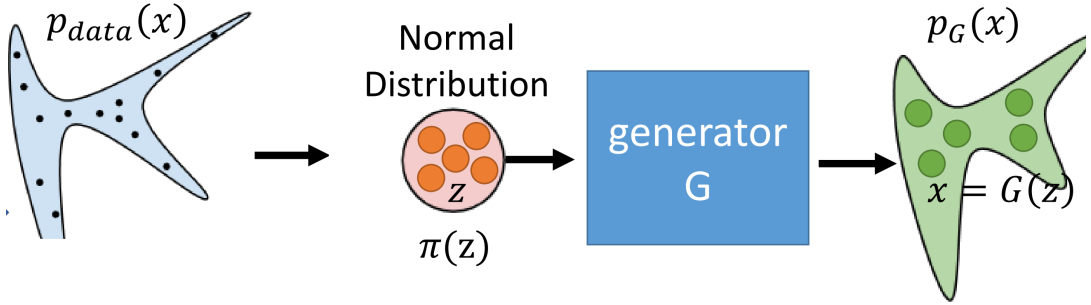
$p(x|z;\theta)$



Posterior density of z:

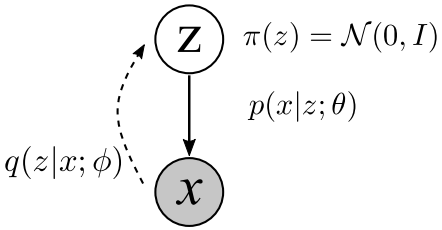
$$P(z|x_i) = \frac{P(x_i|z)P(z)}{P(x_i)} = \frac{p(x_i|z;\theta)\pi(z)}{P(x_i)}$$

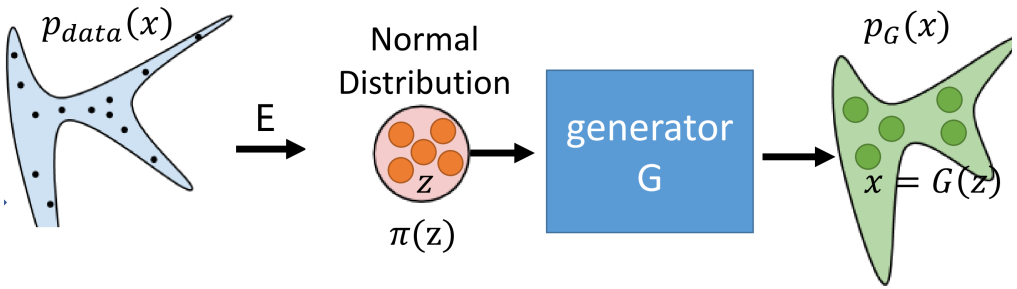
However, data likelihood:

$$P(x_i) = \int p(x_i|z;\theta)\pi(z)dz$$

Intractable

Estimated posterior of latent z





Represent $p(z|x)$ with a encoder E: $q(z|x;\phi)$

The KL divergence between estimated posterior $q(z|x;\phi)$ and true posterior $P(z|x)$:

$$KL\left(q\left(z|x;\phi\right)||P\left(z|x\right)\right)$$

The KL divergence

$$KL\left(q\left(z|x;\phi\right)||P\left(z|x\right)\right) \tag{1}$$

$$= \mathbb{E}_{z\sim q(z|x;\phi)}[\log \frac{q(z|x;\phi)}{P(z|x)}] \tag{2}$$

$$= \mathbb{E}_{z\sim q(z|x;\phi)}[\log q(z|x;\phi) - \log P(z|x)] \tag{3}$$

$$= \mathbb{E}_{z\sim q(z|x;\phi)}[\log q(z|x;\phi) - \log \frac{P(x|z)P(z)}{P(X)}] \tag{4}$$

$$= \mathbb{E}_{z\sim q(z|x;\phi)}[\log q(z|x;\phi) - \log P(x|z) - \log P(z) + \log P(x)] \tag{5}$$

$$\log P(x) - KL(q(z|x;\phi)||P(z|x)) \tag{6}$$

$$= \mathbb{E}_{z\sim q(z|x;\phi)}[\log P(x|z) - \log q(z|x;\phi) + \log P(z)] \tag{7}$$

$$= \mathbb{E}_{z\sim q(z|x;\phi)}[\log P(x|z)] - KL(q(z|x;\phi)||P(z)) \tag{8}$$

The objective

- Maximize the likelihood

$$\log P(x) \geq \log P(x) - KL(q(z|x;\phi)||P(z|x)) \tag{9}$$

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The objective

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- Assume $q(z|x;\phi) = \mathcal{N}(\mu(x;\phi), \sigma(x;\phi))$. In previous, $P(x|z)$ is estimated with $p(x|z;\theta)$, $P(z) = \mathcal{N}(0, I)$.

The objective

- Maximize the likelihood

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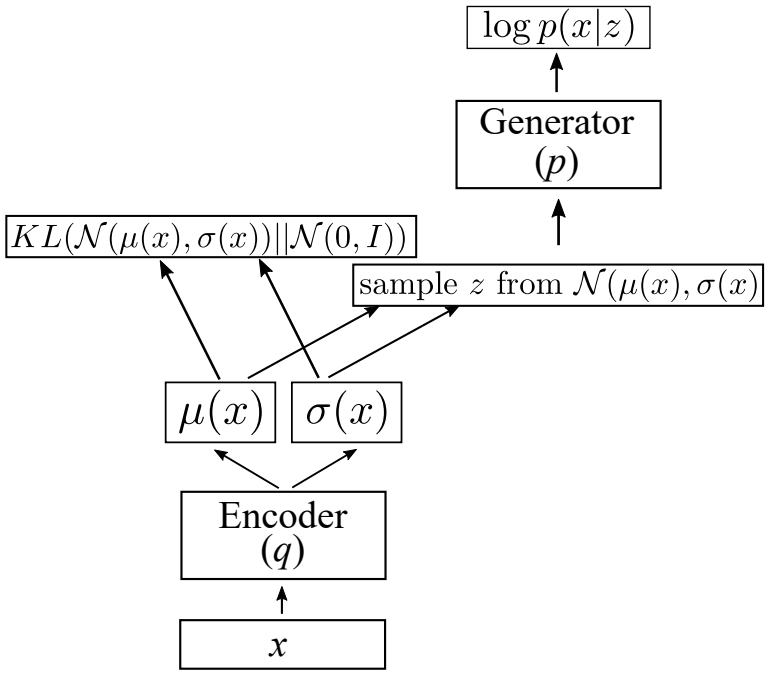
$$\log P(x) \geq \log P(x) - KL(q(z|x;\phi)||P(z|x)) \tag{11}$$
$$= \mathbb{E}_{z\sim q(z|x;\phi)}[\log p(x|z;\theta)] - KL(\mathcal{N}(\mu(x;\phi), \sigma(x;\phi))||\mathcal{N}(0, I)) \tag{12}$$

ELBO

$\mathbb{E}_{z\sim q(z|x;\phi)}[\log p(x|z;\theta)] - KL(\mathcal{N}(\mu(x;\phi), \sigma(x;\phi))||\mathcal{N}(0, I))$ is a **evidence lower bound (ELBO)** of $P(x)$

Reparameterization Trick

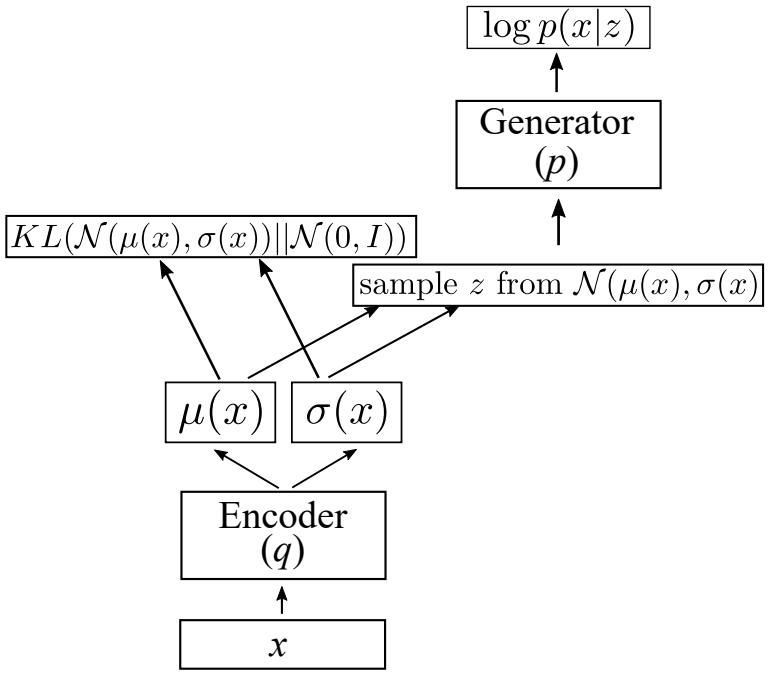
The overall training architecture



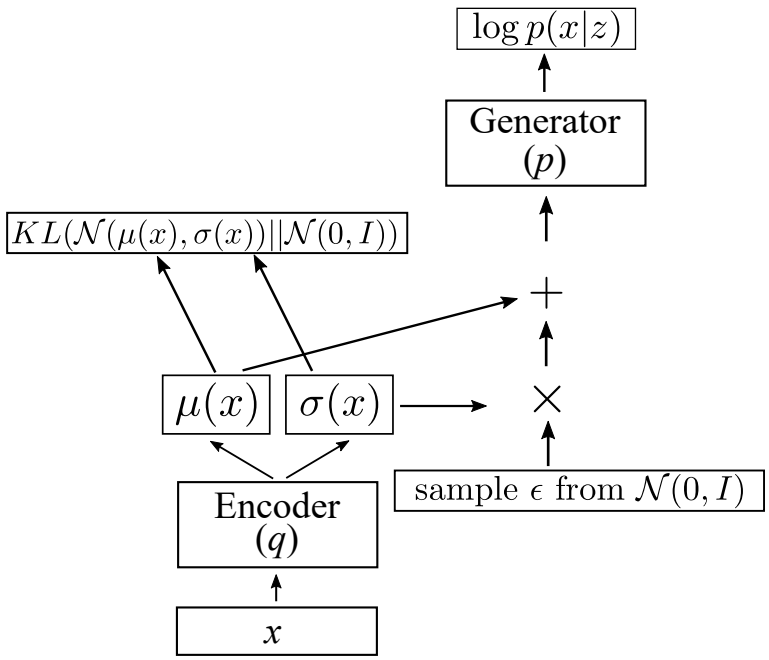
But, the sample procedure is not differentiable!

Reparameterization Trick

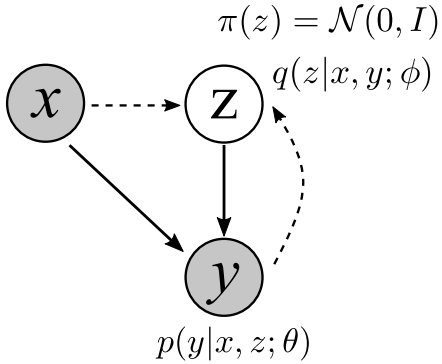
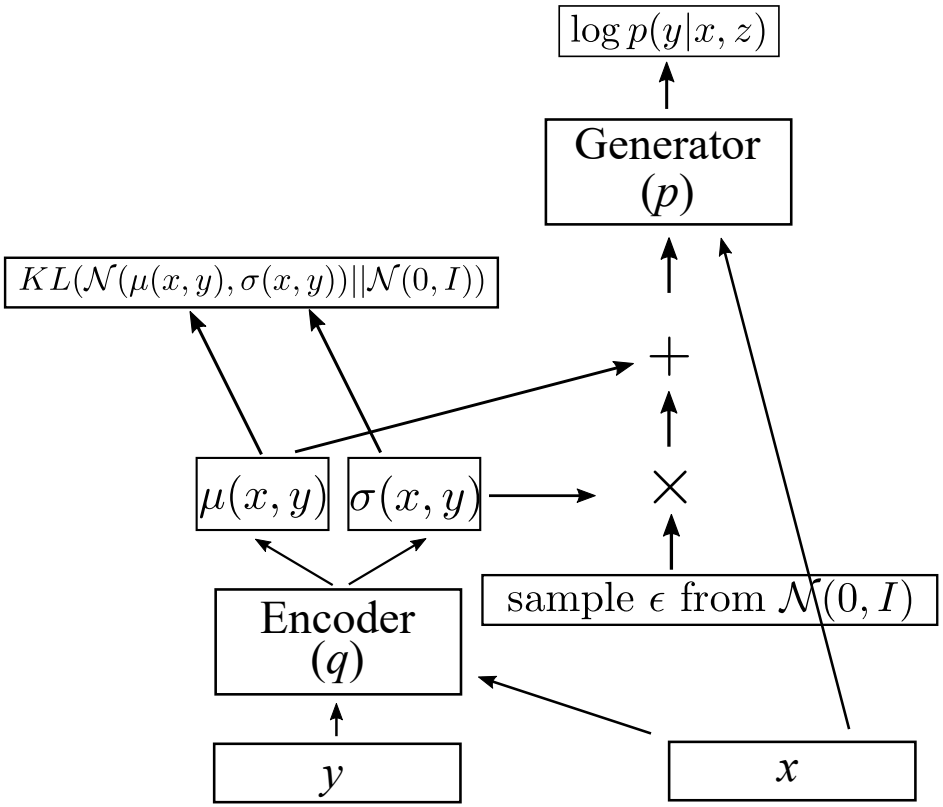
The overall training architecture



Sample from a standard Gaussian



But, the sample procedure is not differentiable!

Conditional VAE

To tackle one-to-many problems, e.g. translation